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HONG KONG

FUDAN UNIVERSITY 復旦大學

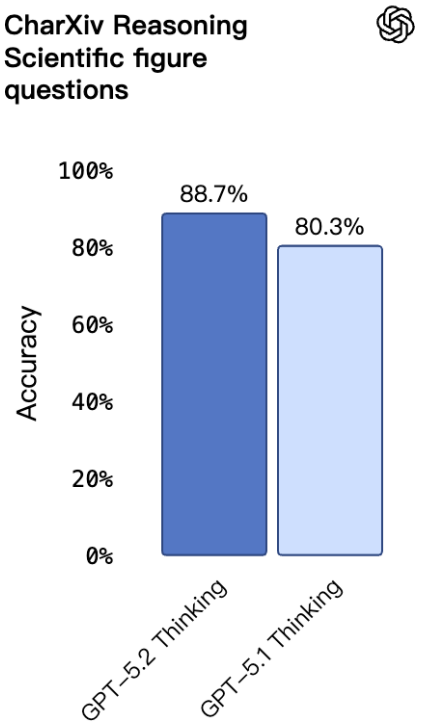


Do Language Model Agents Align with Humans in Rating Visualizations? An Empirical Study

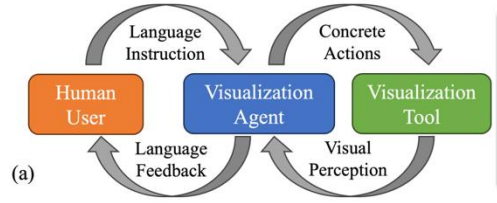
Zekai Shao[*1], Yi Shan[*1], Yixuan He[1], Yuxuan Yao[1], Junhong Wang[1], Xiaolong (Luke) Zhang[3], Yu Zhang[2], and Siming Chen[1]

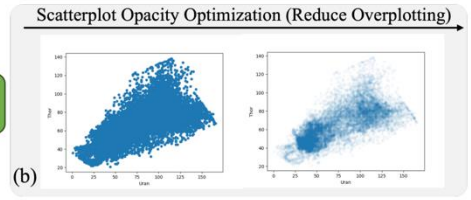
[1] Fudan University [2] University of Oxford [3] Pennsylvania State University

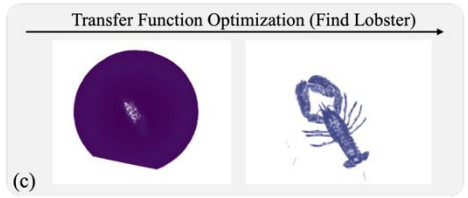
Background: (V)LLM-powered VIS Task

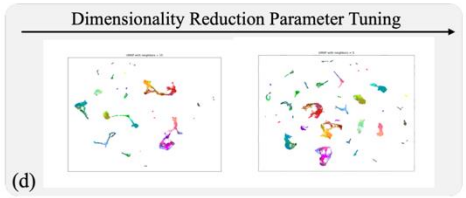


VIS Understanding [1]

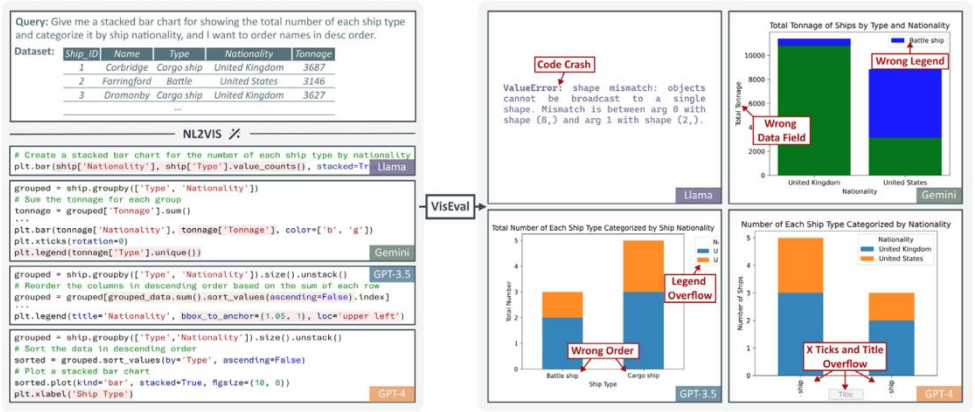








VIS Editing [2]



VIS Generation [3]

[1] https://openai.com/index/introducing-gpt-5-2/. 2025.12.11

[2] Liu, Shusen, et al. "AVA: towards autonomous visualization agents through visual perception-driven decision-making." *CGF*. 2024.

[3] Chen, Nan, et al. Viseval: A benchmark for data visualization in the era of large language models. *IEEE TVCG*, 2024

Background: Human-LLM Alignment in VIS

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DracoGPT: Extracting Visualization Design Preferences from Large Language Models

Huichen Will Wang, Mitchell Gordon, Leilani Battle, and Jeffrey Heer

How Aligned are Human Chart Takeaways and LLM Predictions? A Case Study on Bar Charts with Varying Layouts

Huichen Will Wang, Jane Hoffswell, Sao Myat Thazin Thane, Victor S. Bursztyn, and Cindy Xiong Bearfield

Do LLMs Have Visualization Literacy? An Evaluation on Modified Visualizations to Test Generalization in Data Interpretation

Jiayi Hong, Christian Seto, Arlen Fan, and Ross Maciejewski

An Empirical Evaluation of the GPT-4 Multimodal Language Model on Visualization Literacy Tasks

Alexander Bendeck and John Stasko

TABLE 1. Summary of key literature in LLM visualization generation (top section) and understanding (bottom section).

Reference	Key Models	Tasks	Takeaways
VisEval[2]	GPT-4, Gemini-Pro, CodeLlama-7B	Chart generation in Python	GPT tends to perform best
NL4DV-LLM[3]	GPT-4	Vega-Lite generation	GPT outperforms older approaches, but with a slower runtime
ChartGPT[4]	FLAN-T5-XL (fine-tuned)	Vega-Lite generation	Users liked ChartGPT's charts; CoT prompting and fine-tuning help
Bavisitter[5]	GPT-4	Chart linting	GPT can detect and fix common issues in LLM-generated charts
DataTales[6]	GPT-3.5	Writing textual narratives around charts	DataTales was well received by users; LLMs best alongside classic techniques
Chen et al.[7]	GPT-4	Visualization course tasks	Models can write D3.js code, but show atypical patterns of library use
Bendeck and Stasko[8]	GPT-4 V	Visualization literacy tasks	GPT performs well with minimal prompt engineering, but struggles at retrieving exact data values
Hong et al.[9]	GPT-4 V, Gemini-Pro-Vision	Reproduction of VLAT	Model performance depends heavily on the specific chart and analytic task
Dong and Crisan[10]	GPT-4o, Gemini-2.0, ChartGemma, LLaVA-1.5	VLAT, Mini-VLAT	ChartGemma is competitive with GPT; models and humans identify similar regions of importance in charts
Bursztyn et al.[11]	GPT-4 V, T5 (fine-tuned)	Visual explanation generation	Fine-tuned text-only models outperform GPT for explanation generation
Lo and Qu[12]	GPT-4 V, Microsoft Copilot, LLaVA-1.6	Detecting misleading visualizations	Models can detect misleading charts; chain-of-thought prompting is effective
Wang et al.[14]	GPT-4 V, Gemini-1.0	Generating takeaways from charts	LLM takeaways vary by model and chart layout, and differ from those of humans

Summary of Recent Findings [4]

[4] Bendeck, Alexander, et al. "How Visually Literate Are Large Language Models? Reflections on Recent Advances and Future Directions." IEEE computer graphics and applications 45.6 (2025): 120-129.

Our Research Questions

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- RQ1: What are the characteristics of agent ratings?
- RQ2: How well do agent and human ratings align?
- RQ3: How to better align agent and human ratings?

Our Contributions

- Three studies (for RQ1-RQ3) based on existing experiments with human subjects to illustrate the relative alignment between agent ratings and human responses.
- The potential scenarios for agents in swiftly evaluating prototypes and discuss future directions for simulating visualization experiments with ratings.

Study I: What are the characteristics of agent ratings?

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Investigating Time Series Visualisations to Improve the User Experience

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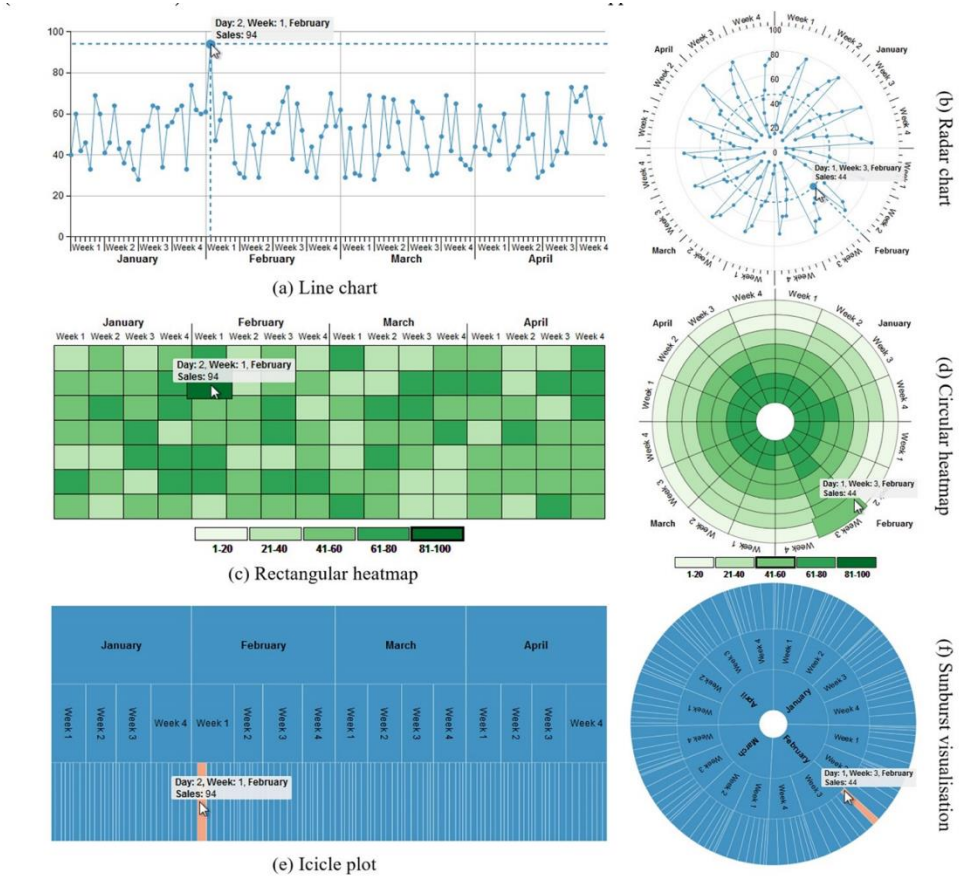
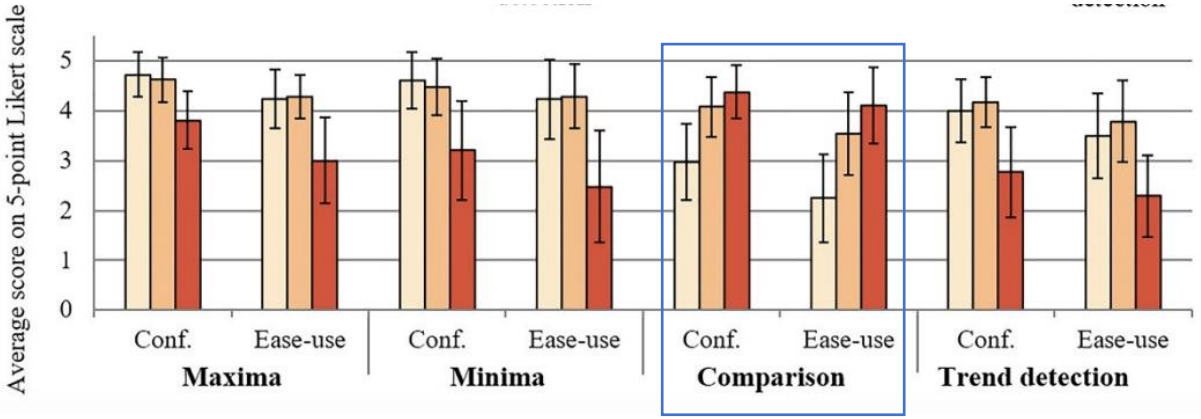


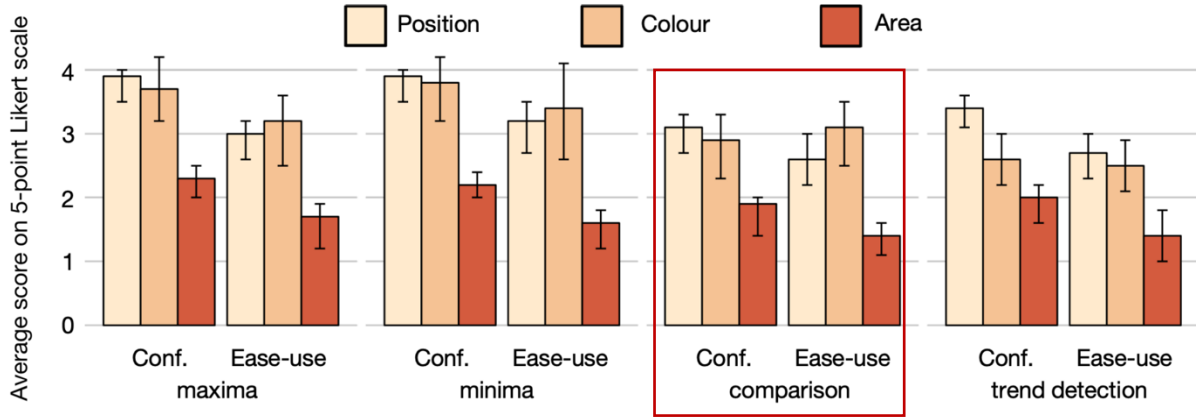
Figure 1. Selected time series visualisations with highlighting and tooltip interaction techniques. Visual encodings: positional (a) & (b); colour (c) & (d); area (e) & (f). Coordinate systems: Cartesian (a), (c) & (e); Polar (b), (d) & (f). Tasks: maxima (a), (c) & (e); trend detection (b), (d) & (f).



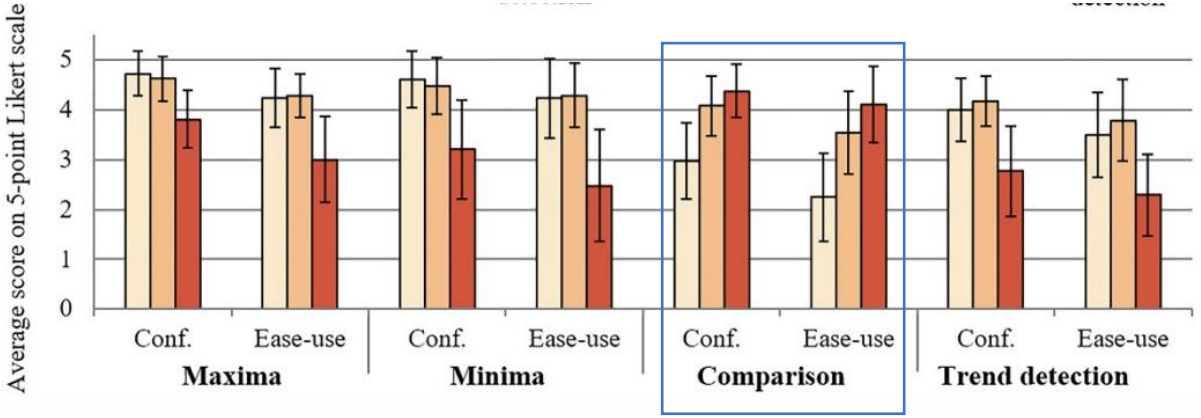
Conclusions in the original paper:

- The area encoding received lower scores in other tasks,
- but excelled in comparison tasks.

✗ (It's opposite in the agent-based study by GPT-4V)



Study I: What are the characteristics of agent ratings?

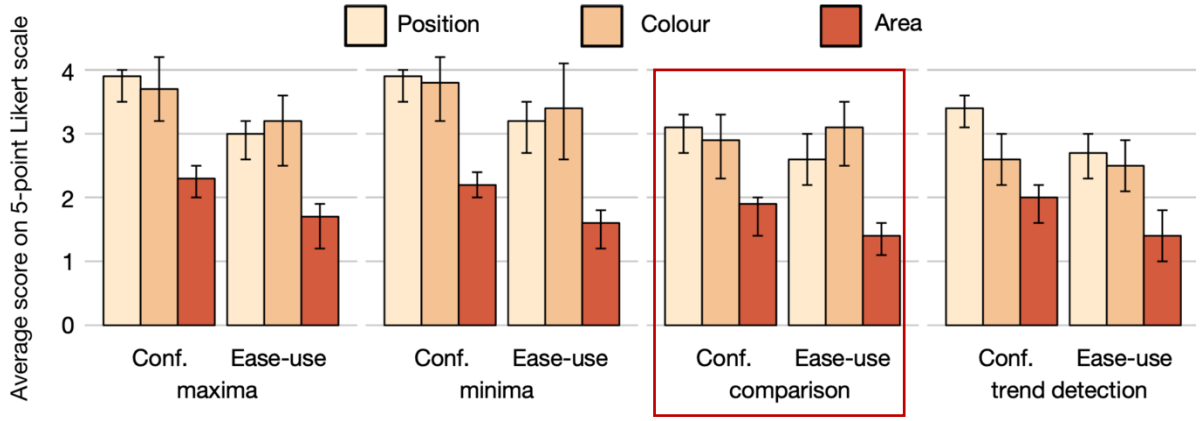


- We remain open to the idea of employing agents for subjective rating.

Conclusions in the original paper:

- The area encoding received lower scores in other tasks,
- but excelled in comparison tasks.

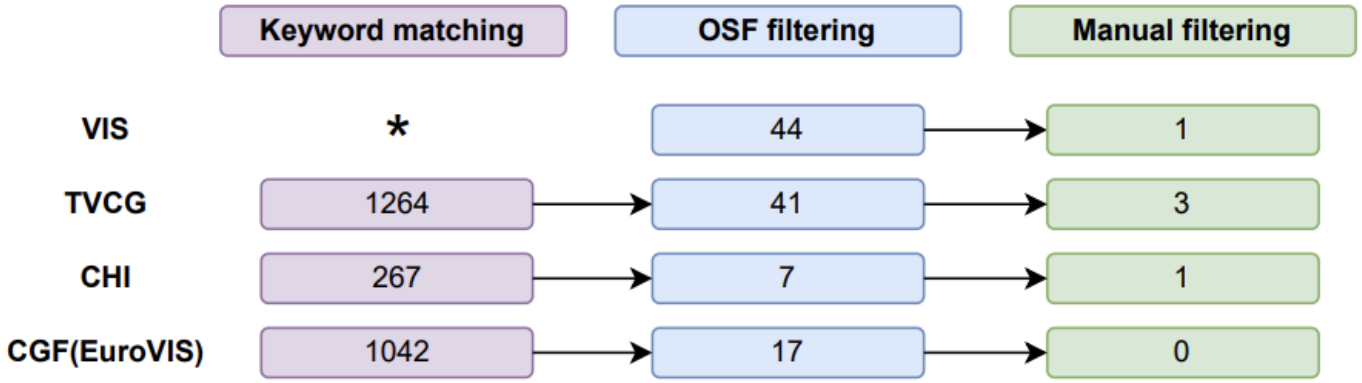
✗ (It's opposite in the agent-based study by GPT-4V)



Study II: How well do agent and human ratings align?

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Paper collection for conducting visualization evaluation with data available in OSF.



We performed agent-based studies on several studies to see Human-LLM alignment.

- 6 empirical papers from VIS, TVCG, and CHI
- Mainly GPT-4V, also a few open-source models (LLaVA-7B/13B, BakLLaVA-7B, etc.)

Study II: How well do agent and human ratings align?

Experiment	Conclusion	Hypothesis	Alignment H-C	Alignment H-A	Alignment C-A
time series [6]	**C1.1**	N	/	/	Y
	C1.2	N	/	/	N
fit estimation [10]	**C2.1**	Y	Y	Y	Y
	C2.2	Y	Y	N	N
	C2.3	Y	Y	P	P
	C2.4	N	/	/	Y
imputation for uncertainty [11]	**C3.1**	Y	P	Y	P
	C3.2	N	N	N	N
timeline [8]	**C4.1**	Y	Y	Y	Y
	C4.2	Y	N	N	P
	C4.3	N	/	/	N
texture [9]	**C5.1**	N	/	/	N
	C5.2	N	/	/	N
	C5.3	Y	P	N	P
	C5.4	Y	P	N	N
magnitude judgement [12]	**C6.1**	Y	Y	N	N
	C6.2	N	/	/	N

- 17 conclusions (C) in six papers;
- Three alignment relations for each between:
  - Hypothesis and C (H-C),
  - Hypothesis and Agents' feedback (H-A),
  - Agents' feedback and C (A-C)
- Three alignment types:
  - Non-alignment (N),
  - Partial alignment (P),
  - Full alignment (Y)

Study II: How well do agent and human ratings align?

Experiment	Conclusion	Confidence	Hypothesis	Alignment H-C	Alignment H-A	Alignment C-A
time series [6]	**C1.1**	H (2.93)	N	/	/	Y
	C1.2	H (2.80)	N	/	/	N
fit estimation [10]	**C2.1**	H (2.60)	Y	Y	Y	Y
	C2.2	M (2.20)	Y	Y	N	N
	C2.3	M (2.20)	Y	Y	P	P
	C2.4	L (1.40)	N	/	/	Y
imputation for uncertainty [11]	**C3.1**	M (1.80)	Y	P	Y	P
	C3.2	L (1.40)	N	N	N	N
timeline [8]	**C4.1**	H (3.00)	Y	Y	Y	Y
	C4.2	M (2.47)	Y	N	N	P
	C4.3	L (1.42)	N	/	/	N
texture [9]	**C5.1**	L (1.40)	N	/	/	N
	C5.2	L (1.20)	N	/	/	N
	C5.3	M (1.80)	Y	P	N	P
	C5.4	M (1.90)	Y	P	N	N
magnitude judgement [12]	**C6.1**	H (2.80)	Y	Y	N	N
	C6.2	M (1.60)	N	/	/	N

N=not align; P=partially align; Y=align

We conducted an experts' confidence coding:

- Invited and consulted five external visualization experts.
- Conducted one-on-one interviews
  1. Present task, materials, and background information
  2. Ask for verbally described hypotheses and confidence rating
     - High (3), medium (2), low (1)
- Adjusted and aggregated all experts' coding.

Study II: How well do agent and human ratings align?

Experiment	Conclusion	Confidence	Hypothesis	Alignment		
				H-C	H-A	C-A
time series [6]	**C1.1**	H (2.93)	N	/	/	Y
	C1.2	H (2.80)	N	/	/	N
fit estimation [10]	**C2.1**	H (2.60)	Y	Y	Y	Y
imputation for uncertainty [11]						
timeline [8]	**C4.1**	H (3.00)	Y	Y	Y	Y
texture [9]						
magnitude judgement [12]						

N=not align; P=partially align; Y=align

Finding: the alignment of agent ratings with human data might correlate with expert confidence for hypotheses.

- Often align ← experts have high confidence.

Study II: How well do agent and human ratings align?

Experiment	Conclusion	Confidence	Hypothesis	Alignment H-C	Alignment H-A	Alignment C-A
time series [6]						
fit estimation [10]	**C2.2**	M (2.20)	Y	Y	N	N
	C2.3	M (2.20)	Y	Y	P	P
imputation for uncertainty [11]	**C3.1**	M (1.80)	Y	P	Y	P
timeline [8]	**C4.2**	M (2.47)	Y	N	N	P
texture [9]	**C5.3**	M (1.80)	Y	P	N	P
	C5.4	M (1.90)	Y	P	N	N
magnitude judgement [12]	**C6.2**	M (1.60)	N	/	/	N

N=not align; P=partially align; Y=align

Finding: the alignment of agent ratings with human data might correlate with expert confidence for hypotheses.

- Often align ← experts have high confidence.
- Partially/not align ← experts confidences are moderate and rely on additional knowledge or unreliable common sense.

Study II: How well do agent and human ratings align?

Experiment	Conclusion	Confidence	Hypothesis	Alignment H-C	Alignment H-A	Alignment C-A
time series [6]						
fit estimation [10]	**C2.4**	L (1.40)	N	/	/	Y
imputation for uncertainty [11]	**C3.2**	L (1.40)	N	N	N	N
timeline [8]	**C4.3**	L (1.42)	N	/	/	N
texture [9]	**C5.1**	L (1.40)	N	/	/	N
	C5.2	L (1.20)	N	/	/	N
magnitude judgement [12]						

N=not align; P=partially align; Y=align

Finding: the alignment of agent ratings with human data might correlate with expert confidence for hypotheses.

- Often align ← experts have high confidence.
- Partially/not align ← experts confidences are moderate and rely on additional knowledge or unreliable common sense.
- Often not align ← when experts have low confidence and set no consistent predefined hypothesis.

Study III: How to better align agent and human ratings?

Experiment	Conclusion	Confidence	Hypothesis	Alignment		
				H-C	H-A	C-A
time series [6]	**C1.1**	H (2.93)	N	/	/	Y
	C1.2	H (2.80)	N	/	/	N
fit estimation [10]	**C2.1**	H (2.60)	Y	Y	Y	Y
	C2.2	M (2.20)	Y	Y	N	N
	C2.3	M (2.20)	Y	Y	P	P
	C2.4	L (1.40)	N	/	/	Y
imputation for uncertainty [11]	**C3.1**	M (1.80)	Y	P	Y	P
	C3.2	L (1.40)	N	N	N	N
timeline [8]	**C4.1**	H (3.00)	Y	Y	Y	Y
	C4.2	M (2.47)	Y	N	N	P
	C4.3	L (1.42)	N	/	/	N
texture [9]	**C5.1**	L (1.40)	N	/	/	N
	C5.2	L (1.20)	N	/	/	N
	C5.3	M (1.80)	Y	P	N	P
	C5.4	M (1.90)	Y	P	N	N
magnitude judgement [12]	**C6.1**	H (2.80)	Y	Y	N	N
	C6.2	M (1.60)	N	/	/	N

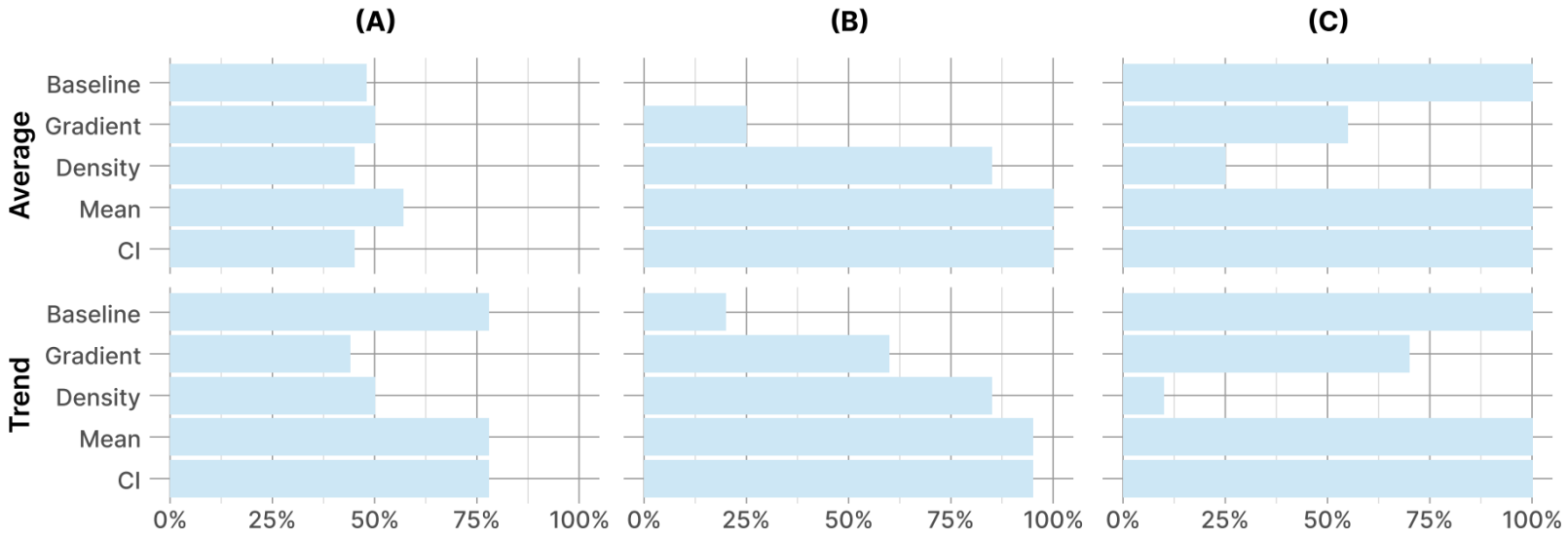
N=not align; P=partially align; Y=align

Trials for improving the alignment:

1. How Do Visual and Textual Cues Influence Reasoning?
2. Does Knowledge Injection Improve Alignment?

Study III: How to better align agent and human ratings?

1. How Do Visual and Textual Cues Influence Reasoning?



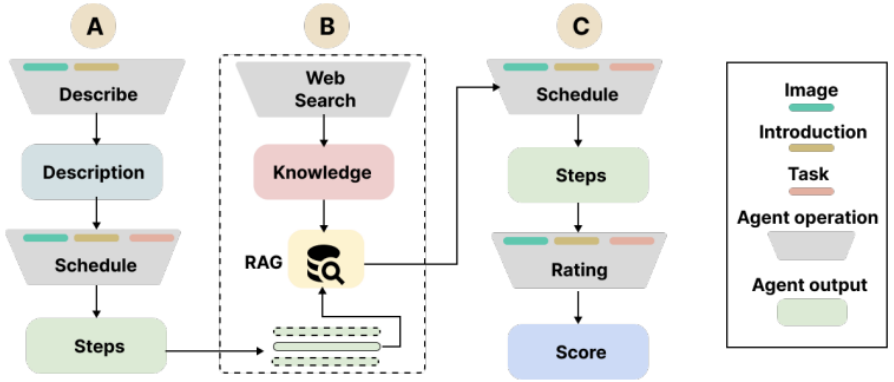
Proportion of agent ratings ≥ 3 for imputation methods addressing uncertainty [5].

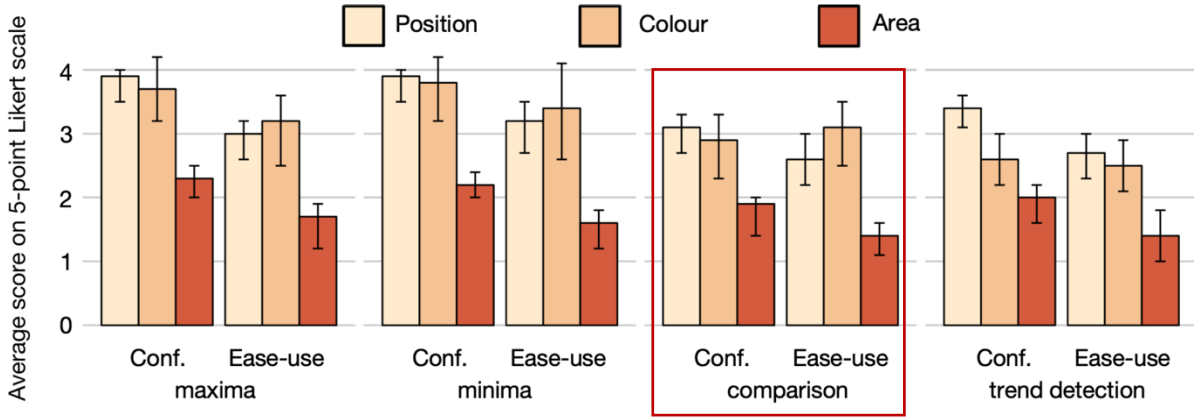
They feature two tasks (estimating the average and trend) across three conditions: (A) text and images combined, (B) text only with the same prompt, and (C) text only with a general description.

[5] Sarma, Abhraneel, et al. "Evaluating the use of uncertainty visualisations for imputations of data missing at random in scatterplots." IEEE Transactions on Visualization and Computer Graphics 29.1 (2022): 602-612.

Study III: How to better align agent and human ratings?

2. Does Knowledge Injection Improve Alignment?





Knowledge injection from web search (B) improve the alignment in comparison tasks of time series visualizations.

The area encodings occasionally achieved the highest scores in RAG-enhanced procedure (19.0% for confidence level; 20.7% for ease of use).

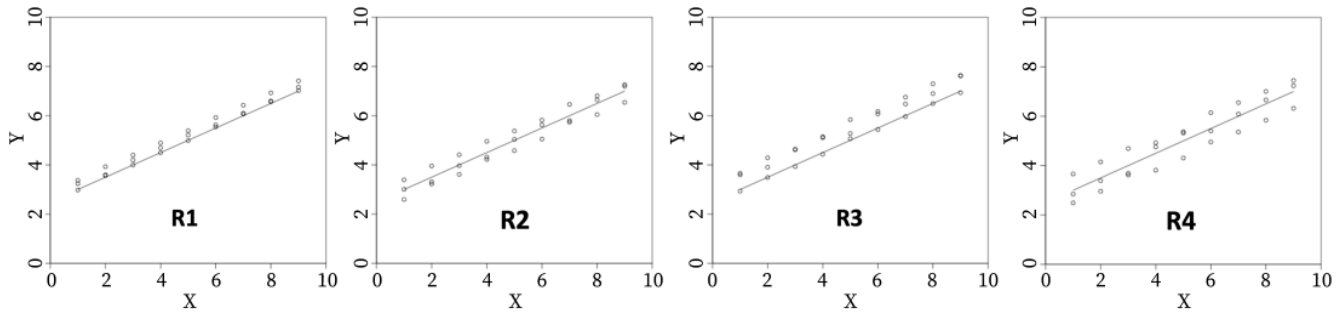
Potential Scenarios: Fast Prototyping Study Design

Motivation:

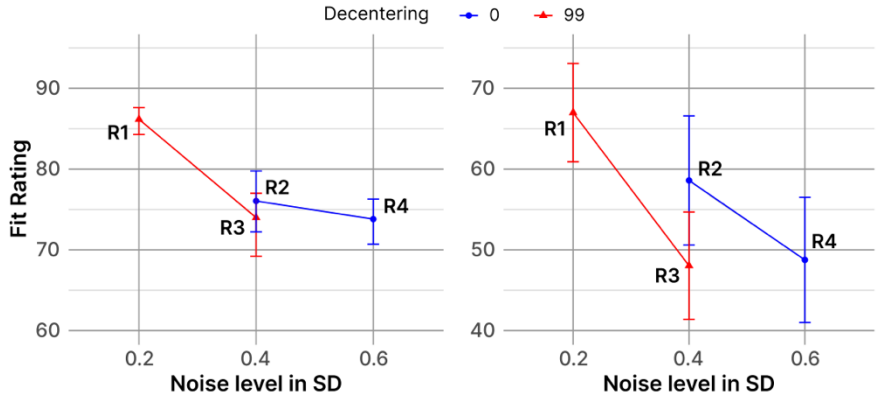
- There's a good alignment for C2.4 in Fit Estimation study.
- Experts have low prior confidence in the study results.
- We wonder if this alignment can be maintained if we employ some extreme parameters.

Experiment	Conclusion	Confidence	Hypothesis	Alignment H-C	Alignment H-A	Alignment C-A
fit estimation [10]	**C2.1**	H (2.60)	Y	Y	Y	Y
	C2.2	M (2.20)	Y	Y	N	N
	C2.3	M (2.20)	Y	Y	P	P
	C2.4	L (1.40)	N	/	/	Y

➔ We conducted both new human-subject and agent-based studies by adjusting parameter settings.



Study materials for our new study:
strong decentering (0.99) with 0.2 noise (R1) and 0.4 noise (R3);
no decentering with 0.4 noise (R2) and 0.6 noise (R4)



Fit ratings by agent (left) and human (right).

Using (V)LLM-based agents in facilitating fast, iterative design of experiments might be promising when their ratings can align with those of prototype experiments.

Discussion

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- Possible misuses.
- Data pollution.
- Keeping Pace with State-of-the-Art LLMs.
- Knowledge-assisted and reliable agent simulation.
- Profile-aware agents for diversity simulation.
- Effective input preprocessing and prompting strategies.
- Simulating broader feedback in broader scenarios.
- Limitation and future work in study scale, expert number, and systematical quantitative analysis.



Thank you for listening!
Happy to take any questions.

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Personal Website: https://zekaishao25.github.io/

Project GitHub: https://github.com/ZekaiShao25/Agents-Ratings-in-VIS-Experiments

Project GitHub

Personal Website